# Choose Inputs and Outputs

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

THE CONNECTED WEEK

## Teach a baseline that deserves a fair comparison.

Supply the full activity catalogue, model boundary cases and distinguish the bin example before the independent room task.

**Keep one habit:** Look for exact inputs, all conditions, correct tie order and actual retained results.

MAIN LESSON · SECTION 1

## Prepare this connected week

> Use one main plan and keep practice separate from the independent first attempt.

### Learning goal and precision

Primary target: Turn the project outcome into precise **input** and **output requirement**s

Use this catalogue and task contract unchanged through the shared project unless a later activity explicitly introduces a versioned change.

The **baseline** is deliberately complete and credible: careful manual checking can satisfy the specification.

The educational aim is a traceable, tested recommender and a defensible comparison, not a compulsory claim that automation wins. Minutes are exact paper-model values and resources are assumed available.

By the end, I can specify valid **input**s, exact **output**s and a manual **baseline**.

## What to prepare

- Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.
- The four-row B1 to B4 catalogue and fictional request cards
- A **baseline** sheet with **input**, **output**, reason and pass or fail columns
- Separate storage for teacher-reserved final cases, kept out of development work

## Main teaching plan

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Compare the cases, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

> **Resources:** Use Find and repair a mistake, Practise together, Run the investigation and Build your understanding; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project, Show what you know, Study a complete case and Try a new case independently. Allow extra time for the full task set. Use a coding lab only when this week actually provides one; Your lesson route is the planning section.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

## Opening retrieval and readiness

Without opening the old lesson, explain one key idea from Find a Real Problem. Give an example and a mistake that the idea helps you avoid.

Look for: Describe a specific user difficulty using supplied **evidence** without inventing its cause

Reference example, not the only acceptable wording: For example: fictional pupils choosing activities in the supplied classroom exercise had to choose again in four of ten requests because the first choice did not fit the time or room conditions.

Build an AI app names a solution or construction task without specifying a user difficulty or **evidence** of need.

Without opening the old lesson, explain one key idea from Verify Important **Output**s. Give an example and a mistake that the idea helps you avoid.

Look for: Separate an answer into claims and connect each claim to relevant **evidence**

Reference example, not the only acceptable wording: A directly stated fact is four rolls, 2 m per roll or twelve 50 cm pieces.

A derived quantity is 8 m starting length, 6 m used or 2 m remaining, supported by the relevant calculation from R. R contains no enjoyment **evidence**, so citing it does not support everyone enjoying the activity.

Without opening the old lesson, explain one key idea from Privacy by Choosing Less. Give an example and a mistake that the idea helps you avoid.

Look for: Justify which information a stated task needs and remove unnecessary fields

Reference example, not the only acceptable wording: Keep topic choice; remove name, full date of birth, address and photograph. The totals do not require identity or contact details.

Collecting extra fields creates copies and exposure to manage even if those fields are later ignored or deleted.

It may miss **compatible** options on every request where one exists. Check useful suggestions and whether no-match results are justified, as well as in**compatible** suggestions.

▼ **Earlier wording, alternatives and picture transcripts**

**Teach the learning cycle**

**Retrieval answer guidance**

**Find a Real Problem**

**Verify Important Outputs**

**Privacy by Choosing Less**

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

**What the learner must demonstrate**

Design a paper/plastic sorter that receives an object description. Include a mixed paper-and-plastic cup. Specify **input**s, **output**s and the uncertain-case action. Is the owner's name useful?

### Concept to explain

A useful design makes required information and promised result explicit.

| Term | Meaning |
|---|---|
| required input | information necessary for the process |
| optional input | information that may help but is not required |
| output contract | specific result the system promises to produce |
| missing-data behavior | what happens when an input is absent |

### Prepare the class

This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

### Preparation for the connected explanation

### Readiness answer

### End goal for the pictorial case

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

---

MAIN LESSON · SECTION 2

## Teach the complete contract before a fair baseline comparison

> Provide the full activity catalogue and distinguish it from the two-bin worked case.

## Define the activity contract

Last week we identified a difficulty: some fictional activity choices did not fit the available time or room conditions. Now define the task precisely enough that two reviewers can agree whether an **output** is correct.

A **requirement** such as “suggest a good activity” is too vague. State which **input**s are accepted, what counts as **compatible** and what to do when no option fits.

For this paper project, a valid request contains an available time from 0 to 30 whole minutes and a quiet-room **requirement** written Yes or No. Yes means the activity must be marked quiet.

No means either quiet or non-quiet activities are permitted; it does not mean noise is required. A missing or invalid value needs clarification, not a guessed value or a no-match claim.

A **compatible** activity takes no more than the available minutes and meets the quiet-room condition. If several activities fit, suggest the first in catalogue order. If none fits a valid request, report no match.

Give a brief reason. This is a suggestion only: it does not book resources, tell anyone to begin or guarantee enjoyment.

## Read the complete activity catalogue

| Catalogue order and activity | Minutes required | Quiet activity |
|---|---|---|
| B1 Bookmark drawing | 10 | Yes |
| B2 Story circle | 8 | No |
| B3 Pattern puzzle | 12 | Yes |
| B4 Tower challenge | 15 | No |

These four rows are the complete approved catalogue for the paper model. Treat their times as fixed task values, not measured guarantees about real pupils. All required materials are assumed available.

The catalogue order B1, B2, B3, B4 is the **tie rule**; it is not a claim that B1 is everyone's favourite.

## Model equality, conditions and missingness

For 10 minutes with quiet required, B1 fits because 10 is no more than 10 and B1 is quiet. B2 fails the quiet **requirement**. B3 and B4 take too long. For 8 minutes with quiet not required, B2 fits; B1 takes too long.

A request for 7 minutes has no **compatible** activity in this catalogue, even when quiet is not required.

Check each condition. A fast activity can still fail the quiet **requirement**. A quiet activity can still take too long. Exact equality is allowed: an activity listed as 10 minutes fits 10 available minutes.

A blank time is different from a valid time of zero.

## Run and record the manual baseline

### The manual checklist baseline

| STAGE | MANUAL ACTION | RECORDED RESULT |
| --- | --- | --- |
| Validate the request | Check minutes and Yes or No | Clarify invalid input |
| Inspect catalogue in order | Check time and quiet for each | Mark compatible rows |
| Choose or explain absence | First compatible row or none | Suggestion or no match |
| Keep the evidence | Record input, result, reason | Comparable baseline record |

The baseline is a complete manual method. A prototype must be compared with this defined procedure, not an invented weak opponent.

Write the request at the top of a sheet. If it is invalid, record what needs clarification and stop. Otherwise mark each catalogue row as **compatible** or in**compatible** using both conditions.

Circle the first **compatible** row in order, or write no match if every row fails. Record the **output** and its reason. This checklist is our **baseline**.

The **baseline** can be correct on all cases when applied carefully. A prototype that matches it has not demonstrated better compatibility.

It may still be easier to use or faster to trace, but those are separate claims that need their own measures. Do not make the **baseline** skip a condition just to make the prototype look better.

## Introduce the separate two-bin contract

A fictional classroom chooses a storage bin for model parts.

Catalogue order: B1 capacity 5, B2 capacity 9. Valid request is a whole part count 1–12.

Return first bin with capacity at least the count; none if neither fits; clarify invalid **input**.

## Return to the activity baseline checklist

AI ACADEMY / AI-1/30 · 03 · INSPECT THE LESSON MODEL

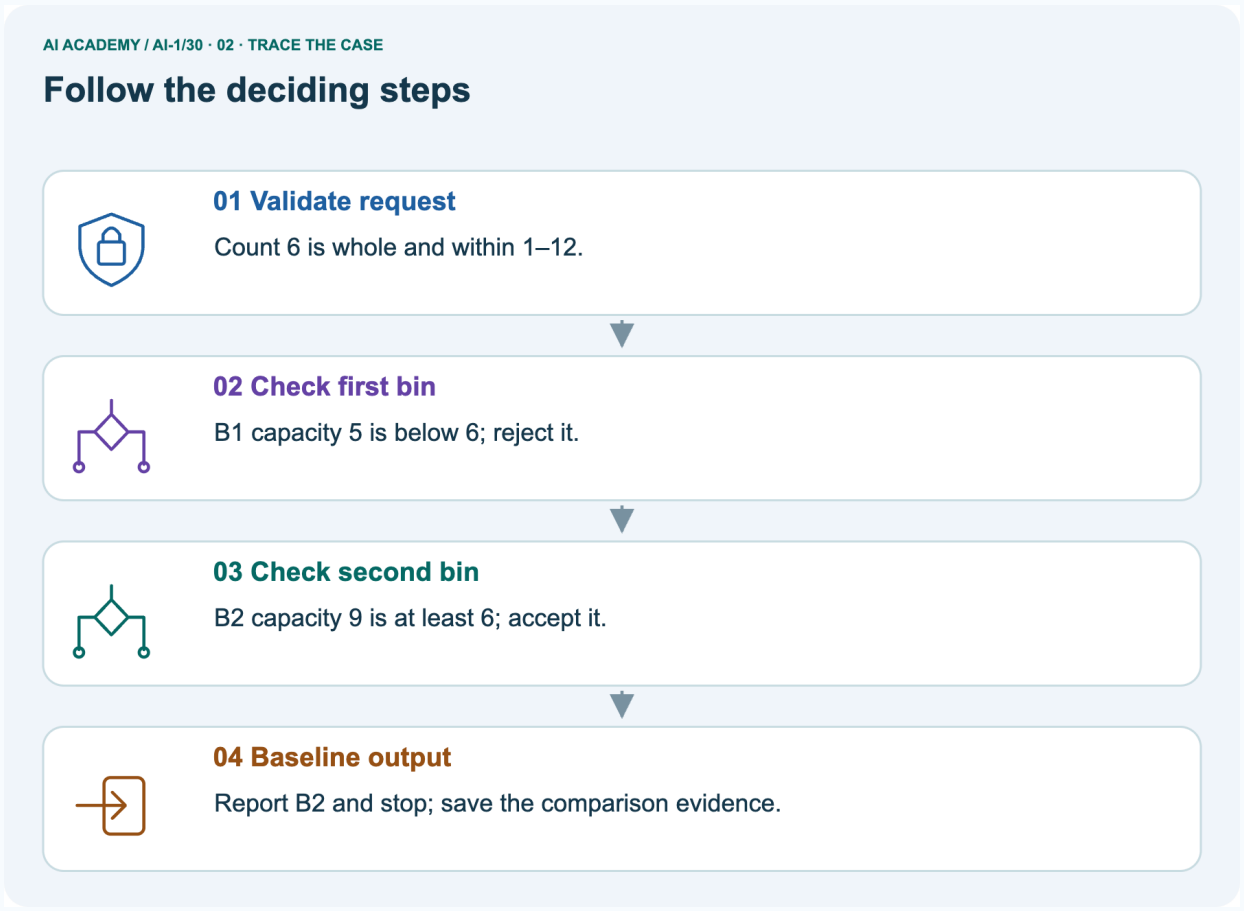
### The manual checklist baseline

| Stage | Manual action | Recorded result |
| --- | --- | --- |
| Validate the request | Check minutes and Yes or No | Clarify invalid input |
| Inspect catalogue in order | Check time and quiet for each | Mark compatible rows |
| Choose or explain absence | First compatible row or none | Suggestion or no match |
| Keep the evidence | Record input, result, reason | Comparable baseline record |

The baseline is a complete manual method. A prototype must be compared with this defined procedure, not an invented weak opponent.

Follow the labelled picture with the matching explanation. Open the vector diagram

## Trace the bin request




Follow the labelled picture with the matching explanation. Open the vector diagram

### Read the picture one step at a time

| STEP | WHAT HAPPENS AND WHY |
| --- | --- |
| 1. Validate request | Count 6 is whole and within 1–12. |
| 2. Check first bin | B1 capacity 5 is below 6; reject it. |
| 3. Check second bin | B2 capacity 9 is at least 6; accept it. |
| 4. Baseline output | Report B2 and stop; save the comparison evidence. |

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

The **baseline** is an ordered catalogue scan, not a preference ranking. It makes expectations testable before a prototype is built.

Count 10 is valid but has no matching bin; that differs from a blank count, which fails the **input** contract. Both situations need explicit **output** behavior.

## Limit bin suitability claims

Bin capacity is the only supplied **fitting** criterion; weight or shape suitability cannot be inferred.

## Change to the exact capacity boundary

Change: Change request 6 to 5.

**Prediction**: The **baseline** returns B1.

Reason: Its capacity equals the request, and at least includes equality.

## Connect the picture to each check

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

### ▼ Earlier wording, alternatives and picture transcripts

Explain the mechanism

Follow the information instead of imagining magic inside the machine.

> **1. Input — information entering**
>
> **INPUT** minimal required information

> **2. Process — the deciding step**
>
> **PROCESS** candidate decision **process**

> **3. Output — the result**
>
> **OUTPUT** specific promised result and uncertainty path

CORE IDEA A useful design makes required information and promised result explicit.

UNDER THE HOOD Interface contracts define **input**s, **output**s, types, allowed values, **missing-data behavior**, and guarantees.

Good design avoids collecting data that cannot improve the promised result and avoids promising outcomes the **process** cannot support.

Draw a fresh example with all three stages:

Every step connects a claim to observable **evidence**.

CASE A sorter can use item type and location found to suggest a bin without collecting a student's name or face.

STEP 1 — OBSERVE minimal required information

STEP 2 — TRACE candidate decision **process**

STEP 3 — CONCLUDE specific promised result and uncertainty path

MODEL REASONING Prefer the design that meets the measurable need with fewer, clearer **input**s and an honest **output**. State how missing values and ambiguous cases are handled before building.

DO NOT OVERCLAIM Every collected field creates cost and risk; every promised **output** needs **evidence**.

What new **evidence** could change this answer?

### Concrete teaching case

A sorter suggests paper or plastic trays using material **observation**s.

Name an **input**, **output** and unnecessary **input**.

### Exact explanation to model

**Input**: material or a suitable image. **Output**: tray suggestion. Home address is unnecessary. Uncertain material needs review.

### Explain the added worked example

### Key terms

| Term | Meaning |
| --- | --- |
| Requirement | A precise condition that the solution must satisfy |
| Baseline | A defined starting method used for comparison |
| Compatible | Meeting all the stated conditions of the request |

| Term | Meaning |
| --- | --- |
| Tie rule | A stated way to choose when several options qualify |
| Acceptance check | A check of whether a result satisfies a requirement |

## Model the reasoning aloud

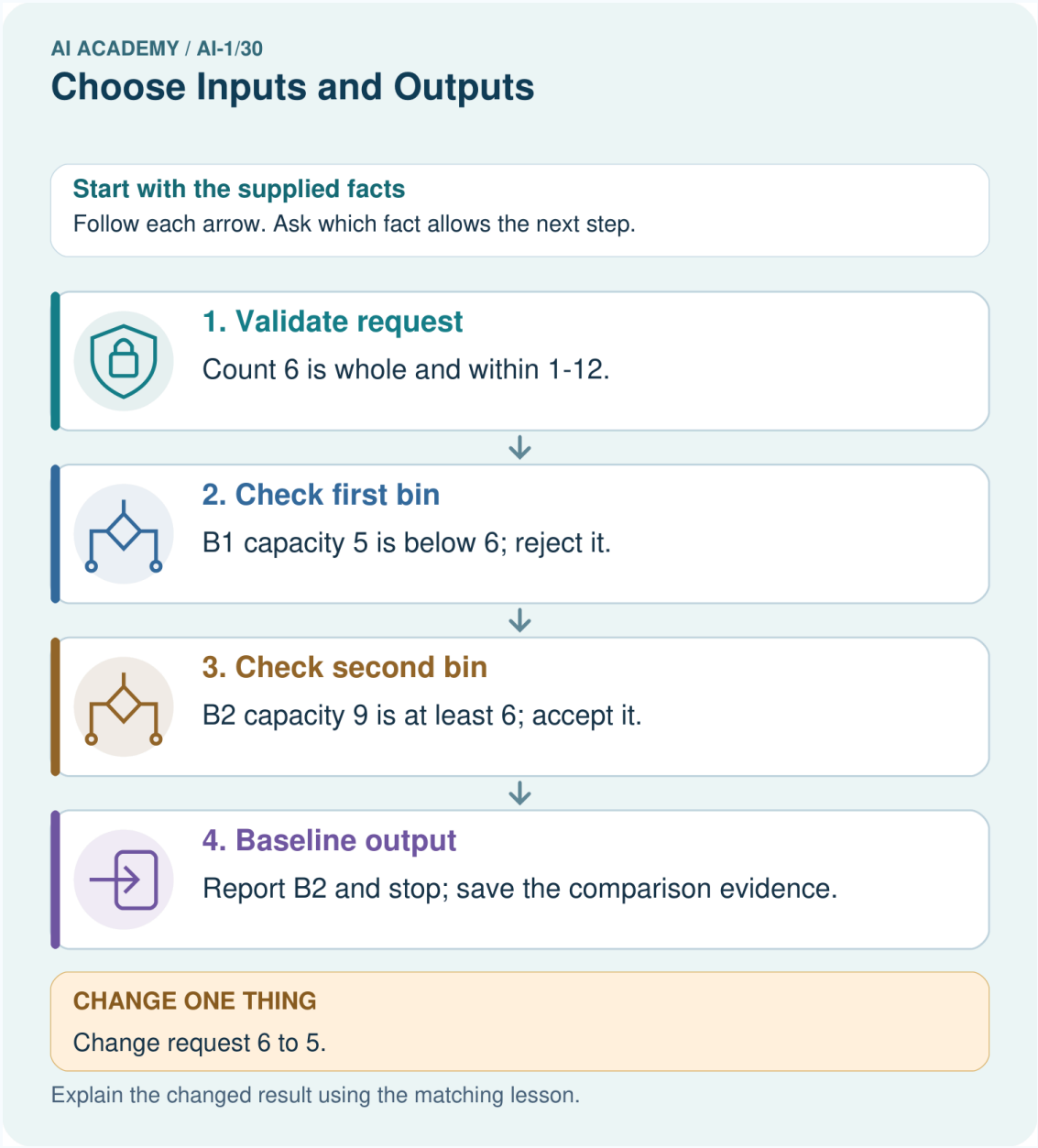
Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

| Thinking move | Teacher prompt |
| --- | --- |
| Plan | What is given? What is the question? Which procedure fits these facts? |
| Do | Point to each input and intermediate step in the worked trace. Name the rule that allows the next step. |
| Check | Does the output answer the question? Which assumption or limitation prevents a larger claim? |

## A pictorial worked case: Choose Inputs and Outputs

### The facts we will use




Follow the labelled picture with the matching explanation. Open the vector diagram

### Why this result follows

### What this example does not establish

### Change one thing in the pictured case

### Ask while showing the picture

MAIN LESSON · SECTION 3

# Guide the reasoning, then remove help

> Collect a response from each learner before discussing the worked key.

### Prompt before revealing

Completion task: Using the catalogue, record the **baseline output** and reason for 10 minutes with quiet required, 8 minutes with quiet not required, and 7 minutes with quiet not required.

Identify which no-match result is justified and why.

## Expected reasoning and fading help

Teacher check: 10 Yes gives B1: 10 fits exactly and it is quiet. 8 No gives B2: B1 is too long, B2 fits and non-quiet is permitted. 7 No gives no match: even the shortest activity requires 8 minutes.

That no-match result is justified by the complete catalogue.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

### Check the misconception before discussion

Use the worked case for **Requirement**s and a simple **baseline**. Proposed explanation: “Quiet required No excludes quiet activities”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: No removes the quiet restriction; it does not reverse it.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

## Use the pictorial check

Trace requests 10 and blank. Why should they not receive the same explanation?

10 is valid but neither bin fits: no match. Blank is missing **input**: request clarification before searching.

Changed-case reasoning: The **baseline** returns B1. Its capacity equals the request, and at least includes equality.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

## Keep each learner explaining

- Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.
- Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.
- In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

▼ **Earlier wording, alternatives and picture transcripts**

Demonstrate and guide practice

### Board sequence

Write the given facts: A sorter suggests paper or plastic trays using material **observation**s.

Write the question to explain: Name an **input**, **output** and unnecessary **input**.

Reveal after **prediction**s: **Input**: material or a suitable image. **Output**: tray suggestion. Home address is unnecessary. Uncertain material needs review.

Use the earlier changed question as optional later practice. Reserve Build a baseline for a different catalogue for the first independent attempt before hints or feedback.

### Classroom dialogue

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

### Model these guided questions

12 Yes: B1 and B3 fit; B2 and B4 are non-quiet, and B4 is also too long. Choose B1 because it is first in catalogue order. 8 Yes: B1 and B3 are too long; B2 is not quiet; B4 fails both. Report no match.

Blank time requires clarification, not a guessed number. Exit: equal successful **output**s on five cases do not show better accuracy for the app.

A separate claim such as faster completion needs recorded times under comparable conditions, with errors and help also recorded.

### Case-specific teaching sequence

| Stage | Teacher action |
| --- | --- |
| Session 1 start 0 to 7 minutes | Retrieve why zero incompatible suggestions is not sufficient evidence of help. |
| 7 to 20 minutes | Define valid input and both compatibility conditions; complete W1. |
| 20 to 33 minutes | Read the complete catalogue and run W2 by hand. |
| 33 to 45 minutes | Complete W3 and distinguish blank input from a valid zero. |
| Session 2 start 0 to 8 minutes | Retrieve the tie rule and the meaning of Quiet required No. |
| 8 to 23 minutes | Complete the guided baseline and discuss a fair comparison plan. |
| 23 to 38 minutes | Collect W4 independently, requiring reasons for each room result. |

| Stage | Teacher action |
| --- | --- |
| 38 to 45 minutes | Complete W5 and the exit; save the contract and baseline for Week 31. |

### Reduce help deliberately

### Individual reasoning check

### Pictorial practice question

### Reasoned teacher answer

---

MAIN LESSON · SECTION 4

# Run the supplied investigation

> Keep expected results, observed paper traces and remaining limits distinguishable.

## Supplied materials and answer guidance

For the two-design paper audit, the name field changes no compatibility result and is unnecessary for this task. Using the activity catalogue: (12,No) returns B1, first among multiple fitting options; (8,Yes) is a valid no match because B2 is not quiet and other activities exceed 8 minutes; (blank,Yes) needs clarification; (0,Yes) is valid but has no match. Preserve the distinction between missing input and no fitting row. Matching correct outputs establishes agreement on these cases, not speed, usability or universal reliability. Provide the full activity catalogue before the lab. Reserve Build a baseline for a different catalogue for the independent first attempt.

## Run and record

1. Complete the task: Sketch two designs and remove an unnecessary **input**.

2. Keep a normal case and an edge case clearly separated.

3. Record what happened—even when it does not match your **prediction**.

4. Explain the result, one limitation, and the next useful test.

## Keep an inspectable record

| Record | What to capture |
| --- | --- |
| Input or evidence | Exact data, instruction or claim |
| Expected result | Answer or safe fallback declared before the run |
| Observed result | Actual result, including errors |
| Explanation | Rule, calculation or evidence explaining the outcome |
| Next test | One justified change and a new prediction |

## Diagnose the first misunderstanding

Quiet required No excludes quiet activities: No removes the quiet restriction; it does not reverse it.

Ten minutes does not fit a ten-minute limit: No more than includes equality.

Blank time means zero: Blank is missing; zero is a valid **input** with no matching activity here.

## Choose support from the response

### Match the support to the first error

| What the response shows | Next teaching action |
| --- | --- |
| Missing or misread facts | Read the case aloud, mark supplied versus unknown facts, and let the learner restate it. |
| Wrong procedure | Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation. |
| Arithmetic or implementation error | Trace intermediate values or lines; preserve the first mismatch and retry that step. |
| Correct result without a reason | Ask for a trace and a changed case before marking independent understanding. |
| An unsupported wider claim | Ask which supplied evidence supports the claim and which further observation would be needed. |

### ▼ Earlier wording, alternatives and picture transcripts

Run the weekly evidence lab

Predict first, test honestly, and explain what the **evidence** means.

State your **prediction** before the result is known.

INDEPENDENCE After support is removed, repeat the proof with a fresh case.

### Make the lab runnable

Use the supplied investigation for shared practice. Keep predictions and observed paper-trace results separate. Reserve Build a baseline for a different catalogue for the individual first attempt before feedback; do not use the reserved case as a paired lab task.

### Diagnose and reteach

CLAIM TO REPAIR A system can safely infer every missing **input**.

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

### Address these additional misconceptions

A **baseline** must be bad to make a comparison useful: Use a credible simple method and accept a tie or a **baseline** win.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

---

MAIN LESSON · SECTION 5

## Assess the selected new case

> Keep this key until after the first attempt; accept valid alternatives with defensible reasoning.

### Selected independent question and key

Selected case: A paper room recommender uses R1 first, capacity 6, quiet Yes; then R2, capacity 10, quiet No. Valid requests give a whole group size from 1 to 12 and Quiet required Yes or No.

A room fits when capacity is at least group size and the quiet condition is met; choose the first **fitting** room. Record **output**s and reasons for 6 people Yes, 8 people No and 8 people Yes.

Explain how a manual **baseline** can be compared fairly with a prototype. No real booking is allowed.

6 Yes gives R1: equality fits and it is quiet. 8 No gives R2: R1 is too small and R2 meets capacity with quiet not required. 8 Yes gives no match: R1 is too small, R2 is not quiet.

Use the same room catalogue, request set and **requirement**s for both methods and keep all results. Compare suggestions only; do not make bookings.

## Judge explanation, trace and limits

| Dimension | Evidence required |
| --- | --- |
| Explanation | Names the applicable idea or procedure and ties it to the supplied facts. |
| Trace and result | Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution. |
| Limits | States the assumptions, missing evidence or conditions under which the result would change. |

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

## Give one usable repair action

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

### W1 Make a requirement checkable

Replace “suggest a good activity” with the time condition, quiet condition and **tie rule** used in this project. Explain why Quiet required No permits B1 as well as non-quiet activities when their times fit.

Time required must be no more than available time; when quiet is required the activity must be quiet. Choose the first **compatible** activity in B1 to B4 order.

No means quiet is not required, not forbidden, so quiet B1 is permitted if 10 minutes fits.

## W2 Run the baseline on three requests

Using the catalogue, record the **baseline output** and reason for 10 minutes with quiet required, 8 minutes with quiet not required, and 7 minutes with quiet not required. Identify which no-match result is justified and why.

10 Yes gives B1: 10 fits exactly and it is quiet. 8 No gives B2: B1 is too long, B2 fits and non-quiet is permitted. 7 No gives no match: even the shortest activity requires 8 minutes.

That no-match result is justified by the complete catalogue.

## W3 Separate invalid input from no matching option

Compare a request with blank minutes and Quiet required Yes to a request with 0 minutes and Quiet required Yes. What should the **baseline** record for each?

Explain why these are different outcomes even though neither produces an activity suggestion.

Blank minutes is invalid and requires clarification of the missing time. Zero is a valid whole number within 0 to 30; no activity takes 0 minutes or less, so no match is justified.

The first lacks a usable **input**; the second has a usable **input** but no **compatible** option.

## W5 Challenge an unfair comparison

Team A tests a prototype on five easy requests and the manual **baseline** on five different requests, then reports only the prototype's successful **output**s. Identify two changes needed for a fair comparison.

If both methods then meet all **requirement**s, what can and cannot be concluded about compatibility?

Use the same requests and catalogue for both with the same judging rules, and retain all **output**s including failures and no-match or clarification results.

If both meet the **requirement**s on those cases, both pass those checks; the results do not demonstrate the prototype has better compatibility or prove either works on every possible case.

## ▼ Earlier wording, alternatives and picture transcripts

Assess and explain the answers

### Worked case answer

**Input**: material or a suitable image. **Output**: tray suggestion. Home address is unnecessary. Uncertain material needs review.

### Additional transfer practice question

A paper sorter must suggest pencil tray or book shelf. Choose one needed **input** and state the **output**. What should happen if the picture is blocked?

### Reasoned practice answer

A clear object picture or relevant shape description can be an **input**; the suggested location is the **output**. A blocked picture should trigger a request for a clearer view or review.

### What counts as evidence

| Criterion | 0 points | 1 point | 2 points |
|---|---|---|---|
| Result | Incorrect or absent | Partly correct | Correct with necessary conditions |
| Reasoning | No supporting trace | Some steps | Clear mechanism, calculation or source support |
| Limit | Overclaims certainty | Vague limitation | Relevant uncertainty or missing fact |
| Transfer | Copies without adapting | Adapts with prompts | Solves a changed case independently |

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

### Feedback

Ask the learner to find the **evidence** that challenges this claim: “A system can safely infer every missing **input**.” Keep the first attempt and an uncoached retry separate.

### Original transfer answer

Use material **evidence** as **input**; **output**s are paper, plastic or review. Send the mixed cup to review under a predefined local rule. Owner name does not determine material.

A description may be incomplete, so the sorter should not claim it has directly inspected the object.

### Explanations for the additional workbook examples

### W1 Make a requirement checkable

### W2 Run the baseline on three requests

### W3 Separate invalid input from no matching option

### W4 Build a baseline for a different catalogue

A paper room recommender uses R1 first, capacity 6, quiet Yes; then R2, capacity 10, quiet No. Valid requests give a whole group size from 1 to 12 and Quiet required Yes or No.

A room fits when capacity is at least group size and the quiet condition is met; choose the first **fitting** room. Record **output**s and reasons for 6 people Yes, 8 people No and 8 people Yes.

Explain how a manual **baseline** can be compared fairly with a prototype. No real booking is allowed.

6 Yes gives R1: equality fits and it is quiet. 8 No gives R2: R1 is too small and R2 meets capacity with quiet not required. 8 Yes gives no match: R1 is too small, R2 is not quiet.

Use the same room catalogue, request set and **requirement**s for both methods and keep all results. Compare suggestions only; do not make bookings.

**W5 Challenge an unfair comparison**

**Assess independent transfer**

**Selected case answer**

**Feedback that the learner uses**

---

MAIN LESSON · SECTION 6

# Connect, adapt and return

> Retain the project evidence and revisit the reasoning on a changed case.

## Project evidence to keep

Freeze the sorter's v1 **input** list, **output contract**, and not-sure behavior.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

Document sorter schema, catalogue order, acceptance rule and all **output** types; use a manual trace as the comparison **baseline**.

## Support while preserving the reasoning goal

Use physical cards, spoken explanations and a three-stage drawing. Read aloud without giving the answer. Stretch: compare two **mechanism**s with the same visible **output** and explain what **evidence** distinguishes them.

Give one tick column for time or capacity and another for quiet. Require both ticks before circling a suggestion. Use a number line to distinguish equal to from greater than.

Extension: explain why a fixed **tie rule** helps reproducibility but does not prove a preference ranking.

W4 requires R1, R2 and no match with capacity and quiet reasons, plus matching **input**s, catalogue and judging rules in the comparison. After support change R2 to quiet Yes and repeat 8 Yes: expected R2.

Ask whether this revises a **requirement** or a catalogue value; it revises the catalogue, so record a new catalogue version and rerun affected checks.

## Plan the delayed check

Suggested return points in this level: ai-1/31, ai-1/33, ai-1/36. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

| Record | Teacher evidence |
|---|---|
| Learner code and date | |
| Concept and case used; prior exposure | |
| Explanation / trace / limits | |
| Support and first repair | |
| Changed case and delayed outcome | |
| Next teaching action | |

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

## Bridge to next week

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Next: Week 31: Design the Decision **Process**. Retrieve today's **evidence** and identify what carries forward and what is new.

Week 31 designs the complete decision **process** using the same **requirement**s and catalogue.

## ▼ Earlier wording, alternatives and picture transcripts

Connect and extend learning

**Helpful Classroom Sorter**

**Support and stretch**

**Exit reflection**

**Bridge to the next week**

**Support and fresh reassessment**

**Plan the delayed return**

**Project evidence from this case**

# Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference: alternative planning checklist**

REFERENCE · SECTION 7

## Reference: alternative planning checklist

> Use as a reminder alongside the main route rather than teaching the cycle twice.

### Preparation and schedule

Teaching target: Turn the project outcome into precise **input** and **output requirement**s

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Compare the cases, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together, Run the investigation and Build your understanding; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project, Show what you know, Study a complete case and Try a new case independently. Allow extra time for the full task set. Use a coding lab only when this week actually provides one; Your lesson route is the planning section.

## Practice and assessment reminders

Before class, solve the supported practice: Using the catalogue, record the **baseline output** and reason for 10 minutes with quiet required, 8 minutes with quiet not required, and 7 minutes with quiet not required.

Identify which no-match result is justified and why.

Reasoned practice answer: 10 Yes gives B1: 10 fits exactly and it is quiet. 8 No gives B2: B1 is too long, B2 fits and non-quiet is permitted. 7 No gives no match: even the shortest activity requires 8 minutes.

That no-match result is justified by the complete catalogue.

Check every learner before discussion: Use the worked case for **Requirement**s and a simple **baseline**. Proposed explanation: “Quiet required No excludes quiet activities”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: No removes the quiet restriction; it does not reverse it.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Change request 6 to 5. Require a **prediction** and a reason before comparing. Expected result: The **baseline** returns B1. Reason: Its capacity equals the request, and at least includes equality.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

| Criterion | Not yet | With support | Independent |
|---|---|---|---|
| Explanation | Names an answer without a deciding fact | Uses a hint to link fact and decision | Explains the deciding fact in own words |
| Trace or test | Cannot connect input and result | Completes steps with prompts | Traces or tests a changed case accurately |
| Limits and revision | Claims more than evidence supports | Names a limit after a prompt | States a defensible limit and useful next test |

Project checkpoint: Prepare the demonstration. Gather reproducible instructions, **evidence**, access checks and remaining limitations.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference: feedback and records**

REFERENCE · SECTION 8

# Reference: feedback and records

> Use digital tools where they help retain an explanation and its evidence.

## Keep practice and independent evidence separate

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

### Repair and portfolio records

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

▼ **Earlier wording, alternatives and picture transcripts**

This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

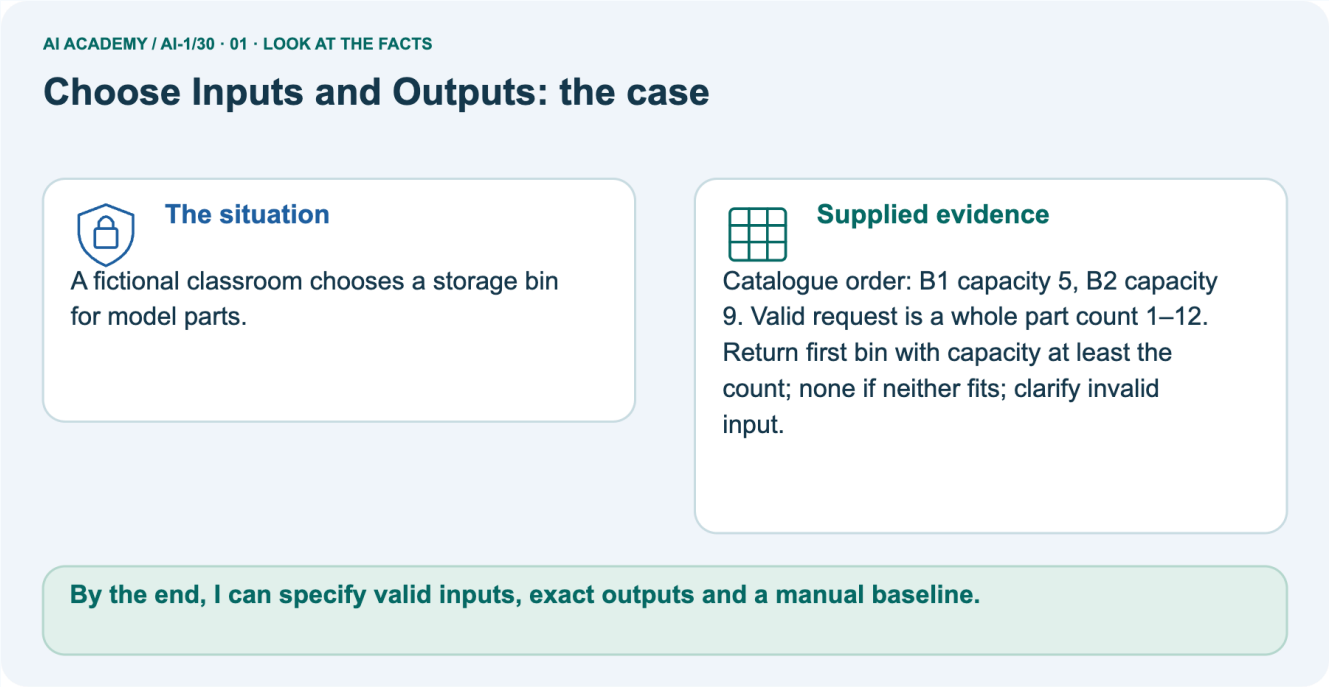
▼ **Reference: teach with the six pictures**

REFERENCE · SECTION 9

## Reference: teach with the six pictures

> Choose the picture for the teaching move and connect it to the explanation.

## 01. Choose Inputs and Outputs: the case




Follow the labelled picture with the matching explanation. Open the vector diagram

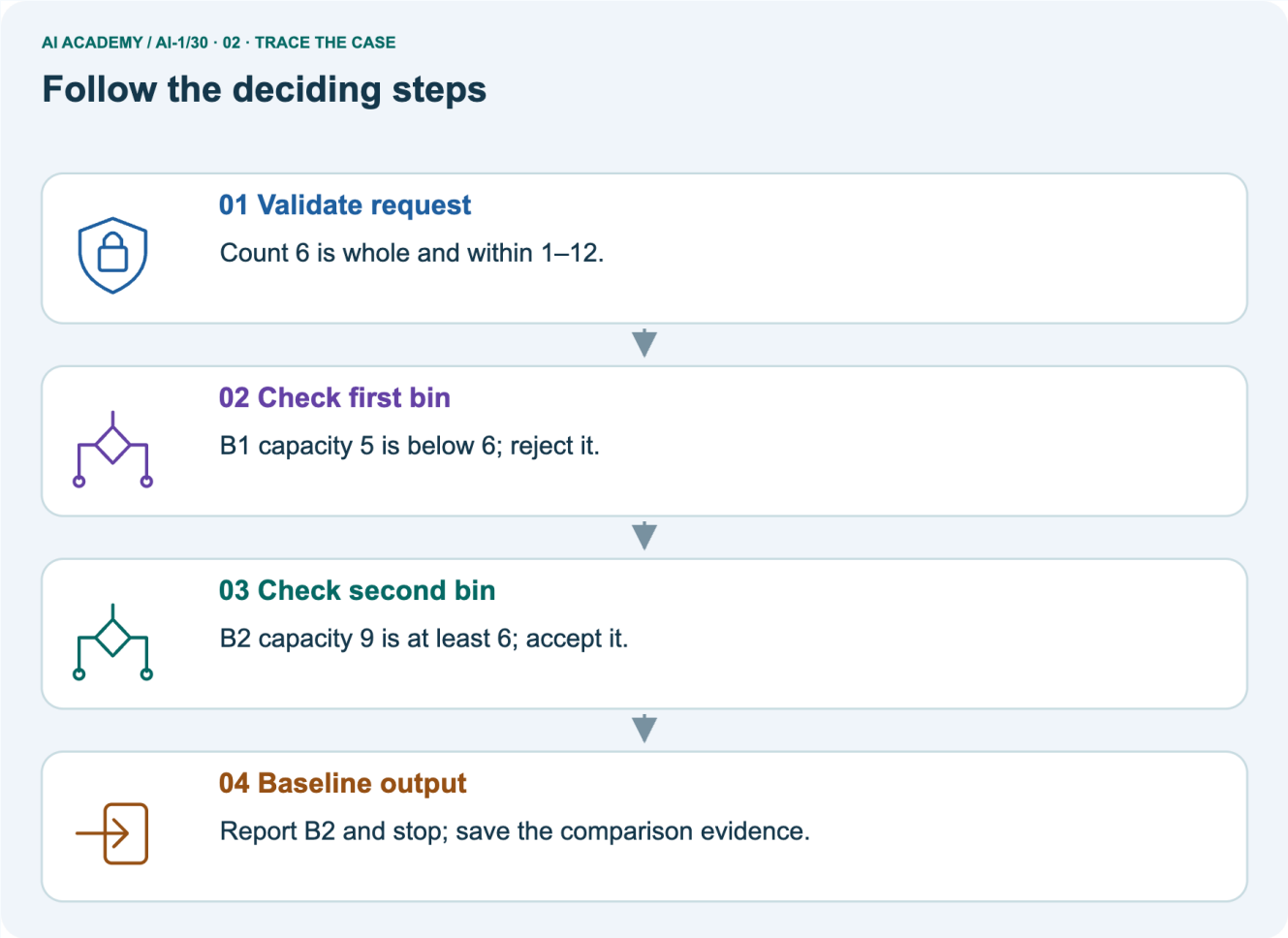
A fictional classroom chooses a storage bin for model parts. Supplied facts: Catalogue order: B1 capacity 5, B2 capacity 9. Valid request is a whole part count 1–12.

Return first bin with capacity at least the count; none if neither fits; clarify invalid **input**.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

## 02. Follow the deciding steps



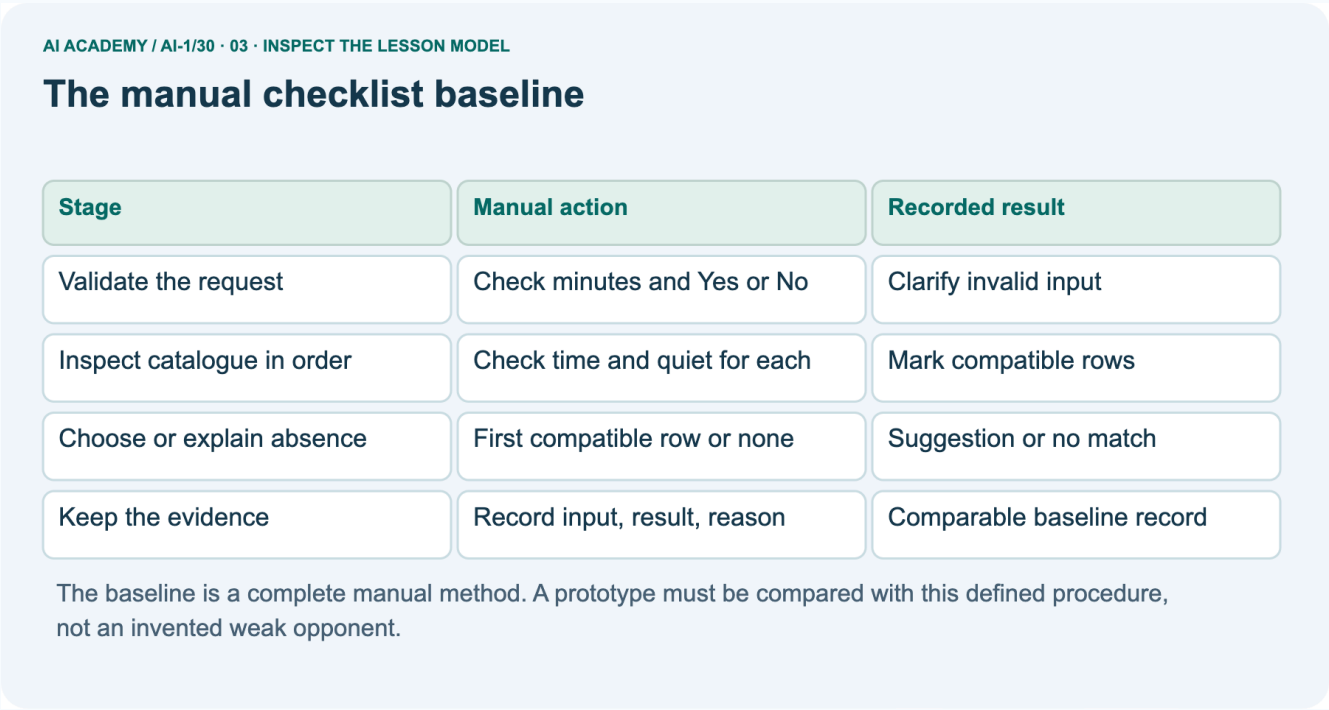

Follow the labelled picture with the matching explanation. Open the vector diagram

Validate request: Count 6 is whole and within 1–12. → Check first bin: B1 capacity 5 is below 6; reject it. → Check second bin: B2 capacity 9 is at least 6; accept it. → **Baseline output**: Report B2 and stop; save the comparison **evidence**.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

## 03. The manual checklist baseline




Follow the labelled picture with the matching explanation. Open the vector diagram

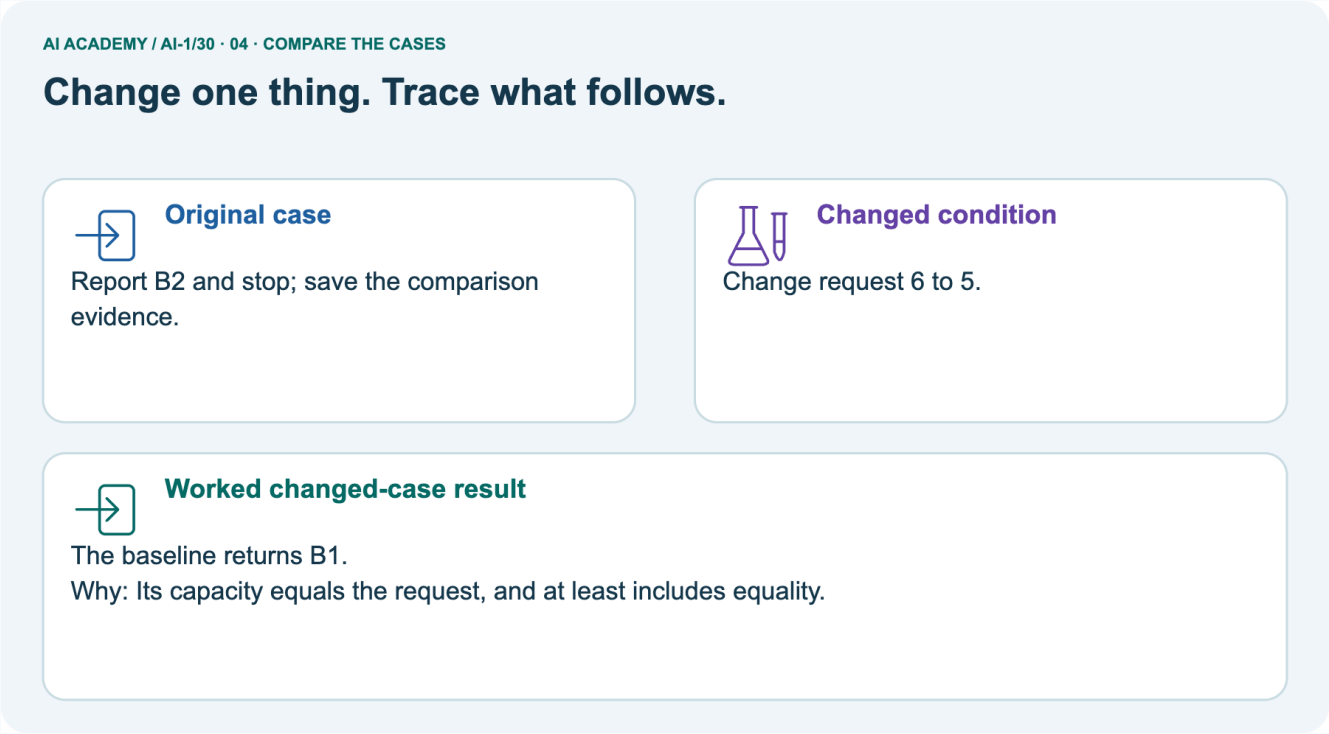
The manual checklist **baseline**. Validate the request | Check minutes and Yes or No | Clarify invalid **input**. Inspect catalogue in order | Check time and quiet for each | Mark **compatible** rows.

Choose or explain absence | First **compatible** row or none | Suggestion or no match. Keep the **evidence** | Record **input**, result, reason | Comparable **baseline** record

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

## 04. Change one thing. Trace what follows.




Follow the labelled picture with the matching explanation. Open the vector diagram

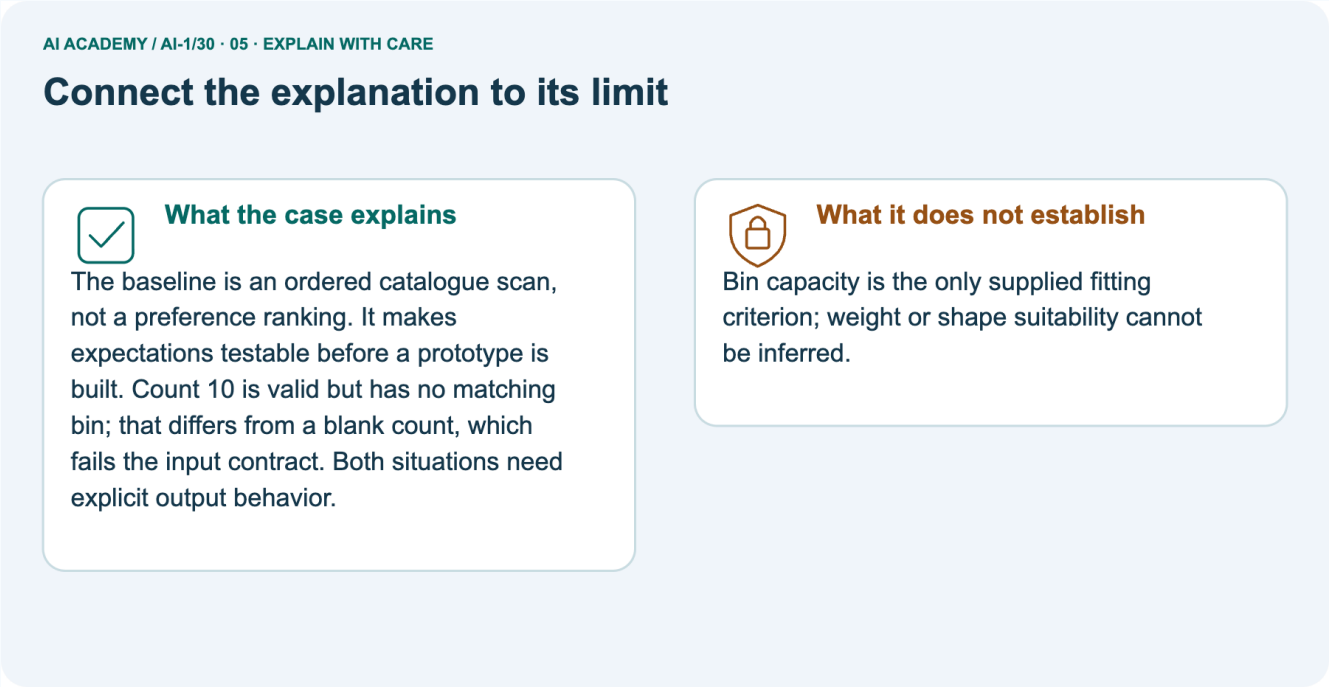
Original case: Report B2 and stop; save the comparison **evidence**. Changed condition: Change request 6 to 5. Expected result: The **baseline** returns B1. Reason: Its capacity equals the request, and at least includes equality.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: Its capacity equals the request, and at least includes equality.

## 05. Connect the explanation to its limit




Follow the labelled picture with the matching explanation. Open the vector diagram

The **baseline** is an ordered catalogue scan, not a preference ranking. It makes expectations testable before a prototype is built.

Count 10 is valid but has no matching bin; that differs from a blank count, which fails the **input** contract. Both situations need explicit **output** behavior.

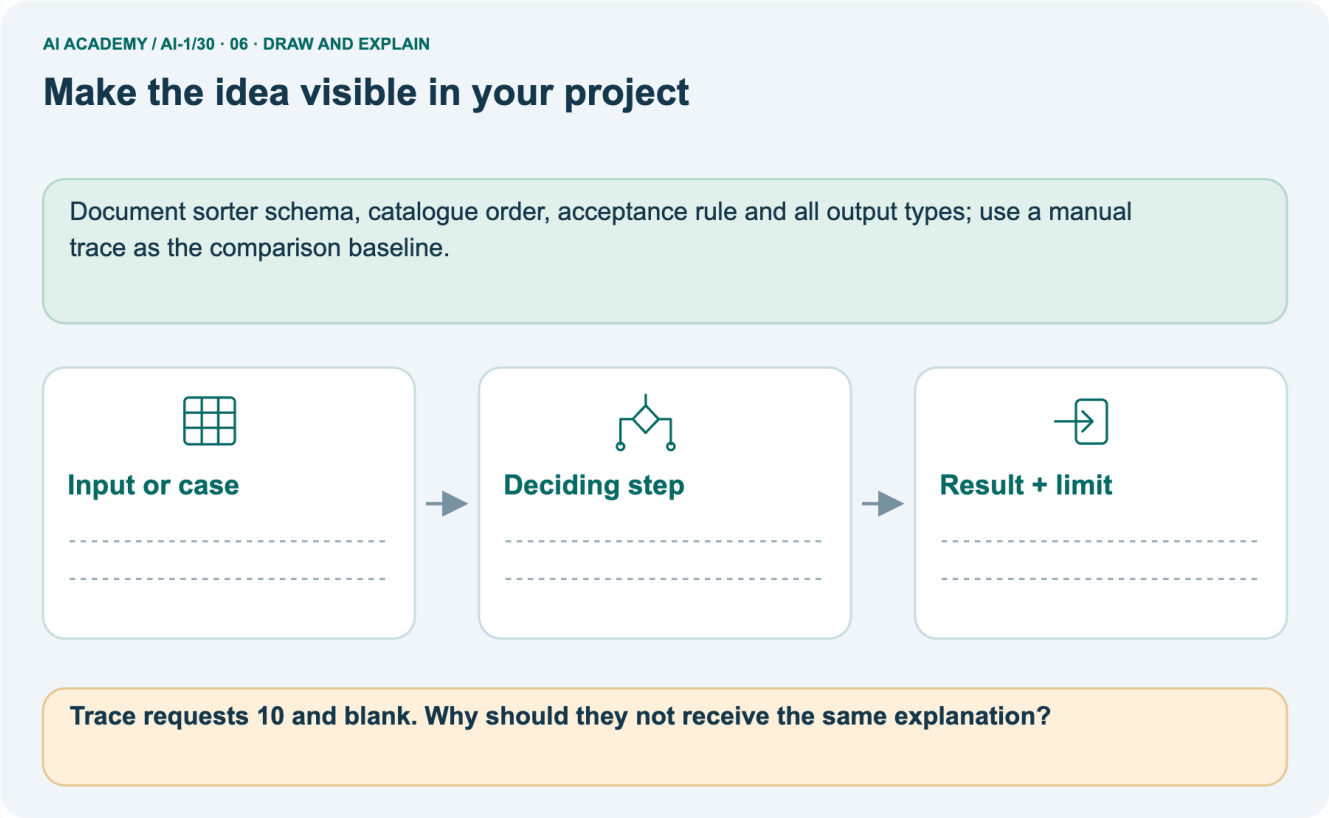
Limit: Bin capacity is the only supplied **fitting** criterion; weight or shape suitability cannot be inferred.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Bin capacity is the only supplied **fitting** criterion; weight or shape suitability cannot be inferred.

## 06. Make the idea visible in your project




Follow the labelled picture with the matching explanation. Open the vector diagram

Document sorter schema, catalogue order, acceptance rule and all **output** types; use a manual trace as the comparison **baseline**. Draw your **input**, deciding step, result and limit. Supported question: Trace requests 10 and blank.

Why should they not receive the same explanation?

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: 10 is valid but neither bin fits: no match. Blank is missing **input**: request clarification before searching.

## ▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

### 01. Choose Inputs and Outputs: the case

### 02. Follow the deciding steps

### 03. The manual checklist baseline

### 04. Change one thing. Trace what follows.

### 05. Connect the explanation to its limit

### 06. Make the idea visible in your project

## ▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.